CAP182 SS2

Part D

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Introduction: STADIOcare patient readmission prediction project

The purpose of this report is to provide recommendations regarding the most suitable model for STADIOcare. STADIOcare needs to reduce operational costs and minimise avoidable hospital readmissions through data driven decision-making. To support this objective, a machine learning project was conducted using the MIMIC-IV clinical data to predict whether a patient would be readmitted within 30 days of discharge. The two models that were developed and evaluated are Logistic regression (model 1) and Random forest (model 2).

1. Best model between model 1 and model 2

Based on the results in Part C, model 1 (Logistic Regression) is the best model for deployment in STADIOcare. The comparison of the two machine learning models showed that Logistic Regression achieved the highest overall Accuracy (78.18%), Precision (44.67%), ROC-AUC (0.700) and Specificity (90.24%). Random Forest achieved a higher Recall (60.99%) and a slightly higher F1 Score (39.18%), indicating that it identified a larger proportion of readmission cases, but at the cost of more false positive predictions.

Recall is important in this project because it measures the proportion of actual readmission cases that were successfully identified by the model. In STADIOcare's healthcare environment, a false negative represents a patient who is at risk of readmission but was not identified by the model, resulting in a missed opportunity for intervention and follow-up care. Whilst Random Forest achieved stronger Recall, it generated more false positive predictions, which would require additional clinical review effort and may reduce operational efficiency.

When considering all evaluation metrics together, Logistic Regression demonstrated stronger overall performance across the majority of measures, particularly Accuracy, Precision, ROC-AUC and Specificity. Logistic Regression therefore provided a more balanced solution by maintaining stronger predictive performance while reducing unnecessary false positive classifications.

The decision to select Logistic Regression was further supported through statistical testing. A McNemar test was performed to determine whether the observed performance difference between the two models was statistically significant. The test produced a statistic of 9508.47 and a p-value of 0.0000, which is substantially below the significance threshold of 0.05. This result indicates that the difference in model performance is statistically significant and is unlikely to be explained by random variation.

Additional confidence was provided through the calculation of 95% confidence intervals for model accuracy. Logistic Regression achieved an accuracy of 78.18% with a confidence interval ranging from 77.93% to 78.42%, while Random Forest achieved an accuracy of 61.51% with a confidence interval ranging from 61.22% to 61.80%. The confidence intervals are narrow and do not overlap, suggesting that the performance difference between the models is both meaningful and consistent.

Based on the combined evidence provided by the evaluation metrics, confidence interval analysis and McNemar statistical test, Logistic Regression is the final model. It achieved superior overall predictive performance, produced fewer false positive predictions and demonstrated statistically significant advantages over Random Forest.

2. How the model will need to be adapted to work with STADIOcare's requested data

The selected Logistic Regression model will need to be adapted to accommodate STADIOcare's operational data environment. Unlike the MIMIC-IV research dataset used during model development, STADIOcare's information is distributed across multiple operational datasets, including admissions, demographics, clinical history, medication records, follow-up information, claims data and procedure information.

Before deployment, these datasets will need to be integrated into a unified analytical environment using common patient and admission identifiers. Additional preprocessing will be required to address missing values, inconsistent category formatting, duplicate records and one-to-many relationships that exist within medication, claims, follow-up and procedure datasets. These preprocessing activities will ensure that the model is trained on reliable and consistent information.

The target variable will remain "Readmitted Within 30 Days" because this aligns directly with the original business objective of reducing avoidable readmissions. However, the model will need to be retrained using STADIOcare's own historical data so that it learns patterns that are specific to STADIOcare's patient population, treatment pathways, medical scheme profiles and operational processes.

Additional feature engineering will also be required. Variables relating to follow-up attendance, medication burden, procedure complexity, claims activity and chronic disease burden have demonstrated potential value within the STADIOcare datasets and should be incorporated into future model development. Expanding these features will allow the model to capture more detailed clinical and behavioural trends associated with readmission risk.

Lastly, the model will also need to account for class imbalance. Analysis of the STADIOcare data indicated that approximately 89% of admissions did not result in

readmission, while only around 11% resulted in readmission. Future development should therefore continue using class-balancing strategies and may benefit from more advanced imbalance-handling techniques such as resampling methods or synthetic minority generation.

3. How the selected model can be improved

Logistic Regression performed very well, however there are a few opportunities to improve its effectiveness. Firstly, additional clinical information should be added into the model. The current model uses patient demographics, admission information, service history, transfer data, and Diagnosis Related Group (DRG) indicators. If we include diagnosis codes, laboratory test results, medication information, and procedure records we would create richer representation of patient data and possibly improve the accuracy of the model's predictions.

Monitoring the model and retraining the model regularly should be implemented. The population of patients, hospital operations, and treatment processes evolve with time. Retraining the model using newly collected data will help maintain predictive accuracy and prevent the model from drifting.

Lastly, more advanced feature engineering techniques could be implemented. Features such as the number of chronic conditions, or frequency of emergency admissions could improve performance of the predictions. These features can provide comprehensive view of a patient's journey and improve the model's predictions.

4. Do the results align with the literature from part A?

The findings align with the literature review in Part A. The research by Almeida et al. found that effective feature engineering and historical healthcare variables improve hospital readmission prediction. This project showed similar outcomes as engineered features like prior admissions and transfer history contributed to strong model performance.

Research conducted by Halac et al. showed that machine learning techniques can be used successfully to predict 30-day readmissions and highlighted the importance of factors such as length of stay. This project also used length of stay information and variables related to admission as the key predictive inputs.

The reviewed literature found that Random Forest and other ensemble methods usually outperform simpler models. In this project Logistic regression performed better than Random forest. A model's performance depends on dataset characteristics, the selected variables, preprocessing approach etc. This reminds us of the importance of evaluating multiple candidate models rather than assuming that a more complex algorithm will automatically produce better results.